Tasks for 03.09.2026 – Advanced Python Programming Lab

1. Using csv
   a. Write a csv file with headers: Student Name, CGPA, Certifications_Count, Internship_Exp_yrs
   b. Add 4 rows of data
   c. Print the location of the file. [Hint: absolute_path = os.path.abspath("students.csv")]
   d. Write the rows one by one
   e. Write the same data as Key Value Pairs using DictWriter and save in students_dict.csv
   f. Write another csv file demonstrating the use of header
   g. Change the mode from w to a and append three more records to the first file
   h. Read the rows from the first csv file and print them all

2. Using Pandas
   a. Load student details to a Dataframe
   b. Use to_csv() to write the the contents from a data frame to csv file
   c. Read from this csv print and print all contents using read_csv()
   d. Use the concatenation method to append few more rows to the csv file
   e. Read the updated file

3. Using json
   a. Create a dictionary of student details as in Question 1
   b. Save the dictionary as a JSON file [Hint: use json.dump()]
   c. Iterate through the JSON file and print the contents
   d. Read a dictionary as a string and convert that into a Python Dictionary object using json.load()
   e. Print the contents of the dictionary

4. Using Pandas
   a. Load a student details dictionary to a dataframe
   b. Use to_json() to save as a json file [hint: use orient]
   c. Read from the json file and print all the contents
   d. Inspect the data using head(), info(), describe() methods
   e. Access the second column and print its contents
   f. Apply filter on a column and print its contents
   g. Add a new column and load the values
   h. Demonstrate updating a value
   i. Write the updated JSON (dataframe) file

Tasks for 07.09.2026 – Advanced Python Programming Lab

5. Using openpyxl
   a. pip install openpyxl
   b. Load an Excel Workbook
   c. Use the ***active*** attribute to call an active worksheet
   d. Count the number of rows and columns in the worksheet
   e. Read data from a specific cell D1

f. Read data from in a row from First column to nth Column
g. Assign a value to a cell using cell name, for eg, 'K1'
h. Assign a value to cell using row and column positions of the cell, for eg, row = 2, column = 10
i. Save the excel file
j. Write a loop to calculate the sum of all values in a row, assign the total value calculated to a cell and name the cell
k. Demonstrate average, countif (count of sales that satisfies a specific condition), sumif, ceiling operations
l. Rename a sheet, delete a sheet, duplicate a sheet
m. Add a bar chart to an Excel sheet:
    i. Load the sheet
    ii. Import Reference to set the minimum column, maximum column, minimum row, max row for the bar chart
    iii. Set the parameters for chart categories
    iv. Create a Bar chart object
    v. Add data
    vi. Set categories
    vii. Set the title
    viii. Add the chart object
    ix. Save the Excel file

6. Using Pandas:
    a. Load the Excel sheet to a data frame
    b. Inspect the data with df.head(), df.info(), df.describe()
    c. Print the data by a specific Column name
    d. Print the data from the first five rows
    e. Filter the data in a specific column with a specific condition
    f. Add a new column from an existing column copying the values that satisfy a specific condition
    g. Update a value
    h. Print the Data Frame
    i. Load the updated Data Frame back to Excel
    j. Print the screenshot of updated Excel sheet

7. NumPy Practice

    a. Create a NumPy Array Object from a List
    b. Create a NumPy Array from a tuple
    c. Create a 3D, 2D, 1D, 0D Numpy Arrays and print the respective size, shape, dimension, datatype
    d. Create 3D, 2D, 1D, 0D Numpy Arrays with all zeros, ones, an integer, random values, linearly spaced, from a specific range
    e. Reshape a 1D array into a 2D, 3D arrays
    f. Reshape a 2D array into a 1D, 2D arrays
    g. Reshape a 3D array into a 2D, 1D arrays
    h. Flatten a 3D and 2D array

8. Slice:
    a. 1D Array:
        i. 2nd element to the last element
        ii. All elements from the beginning to the 4th element
        iii. All alternative elements from the beginning

    b. 2D Array:
        i. Second Row: First element to Third element
        ii. All rows: mid element
        iii. First element of all rows

    c. 3D Array:
        i. Second Element, All rows
        ii. Second Element, Second Row, All Elements

9. Using Array Indexing
    a. Access the 4th element of a 1D Array
    b. Access the Second element, First Row elements of a 3D Array
    c. Add all elements of odd index in a 1D Array
    d. Print the even elements of a 3D Array of 4 elements
    e. Execute the following Code and explain the values of the temp:

```
arr = np.array([[-1, 2, 0, 4],[4, -0.5, 6, 0],[2.6, 0, 7, 8],[3, -7, 4, 2.0]])
temp = arr[[0, 1, 2, 3], [3, 2, 1, 0]]
```

    f. Print the indices of arr, constituting temp
    g. Print the elements that are even in a 2D Array
    h. Investigate the following codes:

```
arr1 = np.array([1, 2, 3, 4])
x = arr1.copy()
arr1[0] = 15
arr2 = np.array([-1,-2, -3, -4])
x = arr2.view()
```

    Make changes to the original arrays and comment on what you find
    i. Add an integer to every element of a 1D, 2D, 3D array
    j. Perform multiplication of two 3D Arrays
    k. Perform multiplication of three 2D Arrays
    l. Iterate a 1D array and print the elements
    m. Iterate a 1D array and print the elements. Observe the following code:

```
for x in arr:
  print (x)
```

    n. Nest the above code as required to print the elements of 2D and 3D arrays. Comment on your findings
    o. Use np.concatenate((arr1, arr2) and print the elements of the concatenated array

p. Explore splitting of an array with np.array_split(array, number_of_splits). Document your findings arrays of different dimensions
q. Explore np.sort(arr) and document your findings on 1D, 2D, 3D arrays

10. Series Practice – Part 1

a. Create a simple series of Student_Names and Programmes they belong with
b. Create a series from a list, tuple, dictionary and a custom index
c. Create two series with CAT I and CAT II marks of students and explore element-wise operations
d. Write conditions to check if data in a series are integers/chars/Not a Number
e. Create a series for Product Names, Prices, Quantities in a retail store
f. Create a Multilevel series with arrays, tuple, product and dataframe
g. Access the data elements using First Level Index and Second Level Index.
h. Access a specific data item using .loc

11. DataFrame Practice – Part II

a. Create sample data with 5 rows and 3 columns, (3rd column of numeric type) and load a dataframe
b. Access and print all rows
c. Access and print from 1st row
d. Access and print till last row
e. Access and print from 2nd to 4th row
f. Retrieve all data of all rows from 2nd column
g. Retrieve all data of 4th row from 1st and 2nd columns
h. Retrieve column 3 data with values greater a specific value (pandas filter)
i. Add a new column of data
j. Update a value [hint: df.at[]]
k. Drop a column [hint: df.drop() - column name and axis]
l. Drop a row [hint: df.drop() - row number and axis]
m. Create a dataframe with missing data [hint: use np.nan]
n. Print null status of all data [hint[df.isnull()]
o. Fill null data with 1000 [hint:df.fillna()]
p. Drop rows with null data [hint: df.dropna()]
q. Create two dataframes with an index and add them [hint: Remember Pandas does not add by position, but only by label]
r. Reindex a dataframe with a fill value and print it

12. Data Wrangling Practice

a. Load the csv file into a dataframe https://drive.google.com/drive/folders/1P789WOxevta0HgLC2siVLLiqne0OuG_9?usp=sharing
b. Determine the number of a missing values in each column
c. Replace the missing values in Marks Column with Average Marks
d. Replace CSE with Computer Science & Engineering

e. Filter the students who score less than 50
f. Print the students list with more than 75% attendance
g. Remove the duplicates and print the dataset
h. Display the number of rows before and after the Duplicates
i. Investigate the data for any one of biases
j. What is a typical Data Wrangling Workflow? Where does the Bias Analysis fit in?